

EE/CprE/SE 4920 STATUS REPORT 4

2/27/26 - 3/12/26

Group Number: sdmay26-40

Project Title: True Force Technologies Arcade Rack

Client: True Force Technologies

Advisor: Matt Post

Team Members/Role:

Jacob Garcia - Frontend Development

Dylan Longlett - Frontend Development, Electronics

Andy Drafahl - Team Manager, Frontend Development

Fadi Masannat - Frontend Development

Sofi Gutierrez - Frontend Development

Parnika Dasgupta - Client Liaison, Frontend Development

Colin Yuska - Electronics

Weekly Summary

The team continued working on various minigames for the arcade rack. At this point, many games are being wrapped up and polished. Music for the games is still in development, and the animation for the Power Pull 3000 is just about ready to be implemented. Several parts for the cabinet are coming in, and will soon be integrated with it. Cabinet repairs and touch-ups are ongoing.

Past Week Accomplishments

Jacob Garcia

Rocket game fuel system added, quality of life changes added and added more easter eggs.

Parnika Dasgupta

I worked on my new game called Pop the water balloon. The game requires 2 players. Each player will take turns lifting, which will keep increasing the size of the water balloon on top of the other player's head. After a certain point, the balloon will pop. Whoever is able to pop the balloon first, is the winner.

Dylan Longlett

Put in sufficient hours in previous weeks, is focusing on other work in the meantime.

Fadi Masannat

I started working on the simple start mode that I planned with our client Ben. It is mostly done but needs to be tested with the client to get feedback and iterate on further. I emphasized on the minimalistic arcade and big number in the center approach as this is the baseline mode that encompasses the full power pull loop and experience without extra graphics or mechanics, just as requested.

Sofi Gutierrez

I removed the pipe systems in the minigame and replaced them with vertical objects moving towards the player. I changed certain visuals in the game and started certain power-ups

Colin Yuska

Got receipt printer ordered, and created a plan to get it implemented as soon as it arrives. In addition also followed up with the client about physical panels for the arcade cabinet, and if there was any help they needed getting this figured out.

Andy Drafahl

Worked on music for the space game, just about finished with it. Communicated back and forth with the animator. The animation is pretty much finished, but some final revisions are being made.

Pending Issues

The animator is making final revisions to the animation, but it should be done soon. The cabinet also needs some repairs.

Individual Contributions

Name	Individual Contributions	Hours This Week	Hours Cumulative
Jacob Garcia	- Rocket game fuel system added - Quality of life changes added - More easter eggs	9	31
Parnika Dasgupta	Made the minigame Pop the balloon	5	11
Dylan Longlett	Good for this week, banking on the <i>many</i> hours he put in weeks prior.	0	37.67
Sofi Gutierrez	Changed the type of obstacles in the game, added new visuals, and "powerups".	5	11
Colin Yuska	Finished making the plan for adding the receipt printer, and now working on the exterior panels of the cabinet.	4	11.25
Fadi Masannat	- Finished the requirements of the simple baseline mode for the arcade machine from client - Finished a first iteration of that mode and is ready for testing	4	14
Andy Drafahl	Finished another song for the space game, finishing the last one. Communicated back and forth with the animator.	8.51	12.01

Plans for the Upcoming Week

Jacob Garcia

Continue work on the rocket game.

Parnika Dasgupta

I will work on making sure that the game is connected to the arcade rack. I have to test it out at the arcade rack. I will try to make my game more aesthetically pleasing as well.

Dylan Longlett

Put in sufficient hours in previous weeks, is focusing on other work in the meantime.

Fadi Masannat

Coming from break I would like to polish and finalize this minimal mode and test it with the client. On top of that start planning on the design documentations and other materials we will need for our final presentation.

Sofi Gutierrez

Coming back from break I hope to have the game basically done with sounds effects, visuals, and game logic. I then want to test/make the game playable with the load sensor.

Colin Yuska

Keep working with the client to get custom-ordered parts for the arcade cabinet. Start inquiring about testing locations.

Andy Drafahl

Will finish the music for the rocket game and hopefully another mini game over break. Will make a mock up of the poster for the presentation over break as well.

Summary of Weekly Advisor Meeting

Matt Post pointed out that there are some changes we need to make to the Arduino/Raspberry Pi, switching from a USB cable to UART Tx-Rx communication. Once we do that, we'll send the pins and the code over to Matt so he can finish up the hat. Matt also pointed out that we need to set up our audio.